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DELHI PUBLIC SCHOOL JANKIPURAM, LUCKNOW

SESSION 2024-2025: ANNUAL EXAMINATION

SAMPLE PAPER

SUBJECT: SCIENCE

TIME: 3Hrs. **M.M.: 80**

Date: 28/1/25 **Name :** Mohd. Mifzan Khan **Class/SecIX** D

General Instructions:

(i) This question paper comprises **39** questions. **All** questions are compulsory.

(ii) This question paper is divided into **five** sections – **A, B, C, D** and **E.**

(iii) **Section A** – Questions No. **1** to **20** are Multiple Choice Questions. Each question carries **1** mark.

(iv) **Section B** – Questions No. **21** to **26** are Very Short Answer type questions. Each question carries **2** marks. Answer to these questions should be in the range of 30 to 50 words.

(v) **Section C** – Questions No. **27** to **33** are Short Answer type questions. Each question carries **3** marks. Answer to these questions should in the range of 50 to 80 words.

(vi) **Section D** – Questions No. **34** to **36** are Long Answer type questions. Each question carries **5** marks. Answer to these questions should be in the range of 80 to 120 words.

(vii) **Section E** – Questions No. **37** to **39** are of 3 source-based/case-based units of assessment carrying **4** marks each with sub-parts.

(viii) There is no overall choice. However, an internal choice has been provided in some sections. Only one of the alternatives has to be attempted in such questions.

SECTION A

Select and write the most appropriate option out of the four options given for each of the questions no. 1 to 20. There is no negative marking for incorrect response.

1 Which of the following statements is incorrect about the state of matter?

(a) The force of attraction between the gas particles is very less.

(b) Plasma consists of super energetic and super excited particles.

(c) The plasma glows with a special colour depending on the nature of the gas.

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(d) Bose-Einstein condensate is formed by heating gas of extremely low density.

2. Which of the following properties does not describe a compound? (a) It is composed of two or more elements (b) It is a pure substance. (c) It cannot be separated into constituents by physical means. (d) It is mixed in any proportion by mass.

3. The formula of Ammonium Sulphate is _____ (a) NH_4SO_4
(b) NH_4SO_2
(c) $(NH_4)_2SO_4$
(d) NH_2SO_4

4. The mass number of the element is _____
a) the sum of the number of electrons and protons
b) the sum of the number of protons and neutrons
c) the number of neutrons
d) the number of protons

5. Which statement does not pertain to the endoplasmic reticulum?
a) It serves as a conduit for protein transportation between the nucleus and cytoplasm.
b) It facilitates the movement of materials within the cytoplasm.
c) It may function as a site for certain cellular biochemical processes.
d) It can serve as the site for energy production.

6. This structure of the plant cell is non-living
a) Nucleus
b) Cell wall
c) Cytoplasm
d) Mitochondrion

7. Find out the incorrect sentence
a) Parenchymatous tissue have inter cellular spaces
b) Collenchymatous tissues are irregularly thickened corners
c) Apical and intercalary meristems are permanent tissues
d) Maristematic tissue in its early-stage lack vacuoles

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8. Intestine absorbs the digested food materials what type of epithelial cells are responsible for that

a) Stratified squamous epithelium
b) Columnar epithelium
c) Glandular epithelium
d) Cuboidal epithelium

9. The horizontal straight line on distance-time graph indicates:

a) increasing velocity
b) decreasing velocity
c) zero velocity
d) constant velocity

10. Suppose a boy is enjoying a ride on a merry-go round which is moving with a constant speed of 10 m/s. It implies that the boy is

a) at rest
b) moving with no acceleration
c) in accelerated motion
d) moving with uniform velocity

11. When the woman rowed a boat, she pushed the water backward with oars and the backward water exerts an equal and opposite push on the boat which makes the boat to move forward. Which principle is followed in this case?

a) Inertia of motion
b) Newton's second law of motion
c) Newton's first law of motion
d) Newton's third law of motion

12. A body of mass 9 kg is lying on a surface of table. Calculate the net force acting on it.

a) 9
b) 0
c) 3
d) 18

13. When a ship floats in seawater:

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a) the weight of water displaced is larger than the ship's weight
b) the weight of water displaced is less than the ship's weight
c) the weight of water displaced is equal to the ship's weight
d) it displaces no water

14. Mass of a man is 75 kg and he has reached the centre of the Earth. The mass and weight of man at the Earth's centre, with radius R, respectively, will be:
a) 75 kg, 75 N
b) 75 kg, 0 N
c) 0 kg, 0 N
d) 75 kg, 735 N

15. The distance which compression or a rarefaction travels per unit of time gives————
a) The density of sound wave
b) Speed of sound
c) Wavelength of sound
d) Frequency of sound

16. Sound waves in air is an example of ————————
a) Longitudinal wave
b) Transverse wave
c) Electromagnetic wave
d) None of the options

For Q Nos. 17 to 20 statements are given- One labelled as Assertion(A) and the other labelled as Reason(R). Select the correct answer to these questions from the codes (a), (b), (c) and (d) as given below:

(a) Both Assertion (A) and Reason(R) are true and Reason(R) is the correct explanation of the Assertion (A)
(b) Both Assertion (A) and Reason(R) are true and Reason(R) is not the correct explanation of the Assertion (A)
(c) Assertion (A) is true but Reason (R) is false
(d) Assertion (A) is false but Reason (R) is true

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17. **Assertion:** The valency of aluminium is 3 and oxygen is 2.
Reason: The chemical formula of aluminium oxide is Al_3O_2.

18. **Assertion :** Vessel and sieve tube both are meant for transport purposes.
Reason : Vessels are lignified.

19. **Assertion :** A winded toy car, when placed on floor, starts moving.
Reason : Toy car has kinetic energy stored in it which facilitates its motion.

20. **Assertion:** Compression and rarefaction involve changes in density and pressure.
Reason: When particles are compressed, density of medium increases and when they are rarefied, density of medium decreases.

SECTION B

21. Name and define the process which occurs when a drop of dettol is added to water.

22. Give any two examples in daily life which are based on Newton's third law of motion. Mention the action and reaction force in each example.

23 What is endocytosis? Why is endocytosis found in animal cells only?

24. Draw a well labelled diagram of Neuron.

OR

Give two differences between cartilage and bone.

25. Water hyacinth floats on water surface? Explain.

OR

Why is epidermis important for the plants?

26. Name the two types of animal feed and write their functions.

OR

Why is organic matter important for crop production?

SECTION C

27. Draw a labelled diagram of chloroplast. Why is it called semiautonomous cell organelle?

28. Describe various components of xylem. Write its function.

29. Identify the dispersed phase and dispersing medium in the following colloids.
(a) Fog (b) Cheese (c) Coloured gemstone

30. What are polyatomic ions? Give two examples.

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31.(a) The atomic number of Al and Cl are 13 and 17, respectively. What will be the number of electrons in Al^{3+} and Cl^{-}?

(b) Define valence electrons. Which electrons of an atom are involved in the chemical bond formation with other atoms?

OR

31(a)Give two important applications of isotopes.

(b)Which isotope of hydrogen is present in heavy water?

32.(a) State the relationship between frequency and time period of a wave.

(b)The wavelength of vibrations produced on the surface of water is 2 cm. If the wave velocity is 16 m/s, find its frequency and time period.

33. (a)Why are ceilings of concert halls and conference halls made curved?

(b) Why are the roofs and the walls of the auditoriums covered with sound absorbent materials?

(c) State two necessary conditions for echo to be heard.

SECTION D

34.(a) To make a saturated solution, 36 g of sodium chloride is dissolved in 100 g of water at 293 K. Find its concentration at this temperature.

(b) Differentiate between homogeneous and heterogeneous mixtures with examples.

35.Draw a plant cell and label the parts which

a) determines the function and development of the cell nucleus
b) packages materials coming from the endoplasmic reticulum Golgi apparatus
c) provides resistance to microbes to withstand hypotonic external media without bursting cell wall
d) Is site for many biochemical reactions necessary to sustain life? nucleus cytoplasm
e) Is a fluid contained inside the nucleus? nucleoplasm

OR

35.(i)Which cellular structures are involved in the following functions:

a) Intracellular digestion
b) Cellular secretion
c) Protein synthesis

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d) Selected entry and exit of substances in an out of cell

(ii) What are genes? Discuss their chemical nature ?Write their function.

36. (a) State Newton's second law of motion and show that the first law of motion can be mathematically stated from the mathematical expression for the second law of motion.

(b) A stone dropped from a window reaches the ground in 0.5 seconds (given $g = 10\ ms^{-2}$).

(i) Calculate the speed just before it hits the ground.

(ii) What is the average speed at t = 0.5 s?

(iii) Calculate the height of window from the ground.

OR

36.(a)Explain the following briefly:

i. A cricket ball causes much severe injury than a tennis ball on hitting a spectator.

ii. An applied unbalanced force causes a change in momentum.

iii. A greater force is required to impart greater velocity to an object.

SECTION E

37. A student was asked by his teacher to verify the law of conservation of mass in the laboratory. He prepared 5% aqueous solutions of NaCl and Na_2SO_4. He mixed 10 mL of both these solutions in a conical flask. He weighed the flask on a balance. He then stirred the flask with a rod and weighed it after sometime. There was no change in mass.
Read this narration and answer the questions given below:

1. Was the student able to verify the law of conservation of mass?
2. If not, what was the mistake committed by him?
3. In your opinion, what he should have done?

OR

3.What is the molar mass of Na_2SO_4? (Given atomic mass: Na=23, S=32, O=16)

38. Read the given passage and answer the questions that follow based on the passage and related studied concepts.
A school picnic was organised at Fateh Sagar Lake, Udaipur. Children were allowed to take boat rides. All types of boats can be seen at the lake like small boats, cruise ships and many more. Shilpa found that boats are floating on water where she can also see instructions on board that "Only 8 people are allowed to go on boat ride".

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(1) How do boats float on water?
(2) Will there be any change in the water level near the boat as people start sitting?
(3) What happens to the boat's buoyant force? Compare the buoyant force acting on boat and water.

OR

(3) A floating boat displaces water weighing 6000 Newton.
(i)What is the buoyant force on the boat?
(ii) What is the weight of the portion of boat immersed in the water?

39.The practice of farming and production of maximum agricultural of maximum agricultural yield to meet the needs to present generation without endangering the resource base of future generation is called sustainable agriculture.

Crops can be classified as field crops, plantation crops and garden crops or there maybe commercial classification of crops -food crops, fodder crops and commercial crops. Presently, techniques for new varieties improvement, crop production improvement and crop protection management have been developed to increase the yields of crop.

Based on your knowledge of topic answer the following question

1. The technique of plant breeding in which two plants having the desired characters are made to cross and develop seeds is called
   a) Hybridization
   b) Selection
   c) Genetic engineering
   d) All of these
2. Rabi crops are
(i) Grown in cold and nearly dry condition during winter season
(ii) Sown in the month of October -November and harvested in March -April
(iii) Rice Maize, groundnut etc are examples of Rabi crops

   (a) (i) only
   (b) (i) and (ii) only
   (c) (i),(ii) and (iii) only
   (d) (i) and (iii) only
3. What is green revolution and how did it happen?

OR

3.Explain any two techniques that can be implemented for new varieties improvement.

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